Northwind Traders Sales Performance Analysis

1.Business problem

Northwind Traders wants to understand its sales performance, identify its most valuable customers and products, evaluate shipping costs, and discover opportunities to improve revenue and operational efficiency.

Project objective

Use SQL to analyse transactional sales data and generate actionable business insights to support management decisions.

2. Dataset Overview

The Northwind Traders dataset contains transactional business data covering customers, orders, products, employees, product categories, and shipping companies.

The dataset consists of seven related tables:

- **Customers:** Customer details and company information.
- **Orders:** Customer orders, order dates, and shipping information.
- **Order Details:** Products purchased, quantities, unit prices, and discounts.
- **Products:** Product names, prices, and category assignments.
- **Categories:** Product classifications.
- **Employees:** Employee information.
- **Shippers:** Shipping company information.

Tools Used: DB Browser for SQLite and SQL.

Analytical Techniques: SQL JOINs, aggregate functions, GROUP BY, HAVING, subqueries, Common Table Expressions (CTEs), and date functions.

The analysis focuses on sales performance, customer spending, product demand, and shipping costs.

3. Business Questions and SQL Analysis

3.1 Sales Performance Analysis

Business Question 1: What is the average order value?

Objective: Determine the average revenue generated per customer order to understand typical transaction value.

SQL Query:

```
SELECT
   ROUND(
      SUM(od.unitPrice * od.quantity * (1 - od.discount))
      / COUNT(DISTINCT o.orderID),
      2
   ) AS averageOrderValue
FROM orders AS o
JOIN order_details AS od
   ON o.orderID = od.orderID;
```

Result: The average order value was approximately 1,525.05 monetary units.

Business Insight: Average order value provides a baseline for monitoring transaction performance. Management can track this metric over time and evaluate strategies such as product bundling and cross-selling to increase spending per order.

Business Question 2: Which month generated the highest sales revenue?

Objective: Identify the strongest sales month and understand monthly revenue patterns.

SQL Query:

```
SELECT
   strftime('%Y-%m', o.orderDate) AS salesMonth
ROUND(
      SUM(od.unitPrice * od.quantity * (1 - od.discount)),
      2
   ) AS monthlyRevenue
FROM orders AS o
```

```
JOIN order_details AS od
    ON o.orderID = od.orderID
GROUP BY strftime('%Y-%m', o.orderDate)
ORDER BY monthlyRevenue DESC;
```

Result: April 2015 recorded the highest monthly sales revenue of approximately 123,798.68 monetary units.

Business Insight: Identifying peak sales periods can help management plan inventory, staffing, and promotional activities. Further analysis is needed to determine whether the peak reflects seasonal demand, large individual orders, or other factors.

3.2 Customer Insights Analysis

Business Question 3: Who are Northwind Traders' highest-spending customers?

Objective: Identify customers who contribute the most sales revenue and explore opportunities for customer retention.

SQL Query:

```
SELECT
    c.companyName,
    ROUND(
        SUM(od.unitPrice * od.quantity * (1 - od.discount)),
        2
    ) AS totalRevenue
FROM customers AS c
JOIN orders AS o
    ON c.customerID = o.customerID
JOIN order_details AS od
    ON o.orderID = od.orderID
GROUP BY c.customerID, c.companyName
```

```
ORDER BY totalRevenue DESC

LIMIT 10;
```

Key Findings:

The three highest-spending customers were:

Customer	Total Revenue
QUICK-Stop	110,277.30
Ernst Handel	104,874.98
Save-a-lot Markets	104,361.95

Business Insight:

These customers represent important revenue-generating accounts. Their purchasing activity highlights the value of maintaining strong relationships with high-spending customers.

Business Recommendation:

Northwind Traders should consider customer retention initiatives, such as personalised account management, targeted offers, and regular customer satisfaction reviews.

However, revenue alone should not determine customer profitability. Management should also consider discounts, service costs, and purchasing frequency before allocating additional resources.

Business Question 4: Which customers spend above the average customer spending level?

Objective: Identify customers whose total purchasing value exceeds the average spending across customers who have placed orders.

SQL Query:

```
WITH customer_totals AS (
    SELECT
        c.customerID,
        c.companyName,
        SUM(
            od.unitPrice * od.quantity * (1 - od.discount)
        ) AS totalRevenue
    FROM customers AS c
    JOIN orders AS o
        ON c.customerID = o.customerID
    JOIN order_details AS od
        ON o.orderID = od.orderID
    GROUP BY c.customerID, c.companyName
)
SELECT
    companyName,
    ROUND(totalRevenue, 2) AS totalRevenue
FROM customer_totals
WHERE totalRevenue > (
    SELECT AVG(totalRevenue)
    FROM customer_totals
)
ORDER BY totalRevenue DESC;
```

Key Finding:

The average spending per purchasing customer was approximately **14,222.39 monetary units**.

The query identifies customers whose spending exceeds this benchmark.

Business Insight:

Comparing individual customer revenue with average customer spending helps distinguish high-value accounts from the broader customer base.

Business Recommendation:

Northwind Traders can use this information to segment customers and develop targeted retention strategies. Management should also investigate purchasing patterns among lower-spending customers to identify opportunities for increased engagement.

Management should review inventory levels and replenishment patterns for high-demand products. Lower-volume products should be assessed using revenue, profit margins, pricing, and customer demand before considering discontinuation.

Business Question 6: Which product generated the lowest sales revenue?

Objective: Identify the product with the lowest revenue contribution after discounts.

SQL Query:

```
SELECT
   p.productName,
   ROUND(
      SUM(od.unitPrice * od.quantity * (1 - od.discount)),
      2
   ) AS totalRevenue
FROM products AS p
JOIN order_details AS od
   ON p.productID = od.productID
GROUP BY p.productID, p.productName
ORDER BY totalRevenue ASC
```

```
LIMIT 1;
```

Key Finding:

Chocolade generated the lowest total sales revenue, approximately **1,368.71 monetary units**.

Business Insight:

Low revenue may result from low selling prices, limited sales quantities, discounts, or a combination of these factors.

Business Recommendation:

Northwind Traders should examine Chocolade's sales trends, profitability, and market demand before deciding whether to adjust pricing, increase promotions, or change inventory levels.

Business Question 7: Are there products that have never been ordered?

Objective: Identify products with no recorded sales activity.

SQL Query:

```
SELECT
    p.productID,
    p.productName
FROM products AS p
LEFT JOIN order_details AS od
    ON p.productID = od.productID
WHERE od.productID IS NULL;
```

Key Finding:

The query returned zero records, indicating that all 77 products had been ordered at least once.

Business Insight:

Every product in the catalogue had some recorded sales activity. However, this does not mean all products performed equally well.

Business Recommendation:

Management should evaluate products using sales frequency, revenue, profit margins, and inventory turnover rather than relying solely on whether a product has been ordered.

3.4 Shipping Performance and Cost Analysis

Business Question 8: Which shipping companies handle the most orders, and how do their average shipping costs compare?

Objective: Evaluate shipping companies based on the number of orders handled and their average freight costs.

SQL Query:

```
SELECT
    s.companyName AS shippingCompany,
    COUNT(o.orderID) AS totalOrders,
    ROUND(AVG(o.freight), 2) AS averageFreightCost
FROM shippers AS s
JOIN orders AS o
    ON s.shipperID = o.shipperID
GROUP BY s.shipperID, s.companyName
ORDER BY totalOrders DESC;
```

Key Findings:

Shipping Company	Orders Handled	Average Freight Cost
United Package	326	86.64
Federal Shipping	255	80.44

Speedy Express	249	65.00

Business Insights:

United Package handled the largest number of orders, while Speedy Express recorded the lowest average freight cost.

However, differences in average freight costs do not necessarily mean one shipping company is more cost-efficient than another. Shipment weight, destination, order size, and delivery requirements may influence shipping expenses.

Business Recommendations:

Northwind Traders should investigate whether lower freight costs can be achieved without compromising delivery quality.

Before changing shipping providers, management should compare delivery times, customer satisfaction, shipment characteristics, and service reliability.

The company could also explore negotiating better rates with frequently used shipping partners.

4. Key Findings and Business Recommendations

4.1 Summary of Key Findings

The Northwind Traders sales analysis revealed several important insights into customer behaviour, product performance, sales trends, and shipping operations.

Analysis Area	Key Finding
Sales Performance	Average order value was 1,525.05 monetary units.
Monthly Revenue	April 2015 generated the highest monthly revenue of 123,798.68 monetary units.

Customer Insights	QUICK-Stop was the highest-spending customer, generating 110,277.30 monetary units.
Customer Spending	Average revenue per purchasing customer was 14,222.39 monetary units.
Product Demand	Camembert Pierrot had the highest sales volume at 1,577 units.
Low-Volume Products	Mishi Kobe Niku recorded the lowest sales volume at 95 units.
Product Revenue	Chocolade generated the lowest revenue at 1,368.71 monetary units.
Product Coverage	All 77 products had at least one recorded sale.
Shipping Performance	United Package handled the most orders (326), while Speedy Express had the lowest average freight cost (65.00 monetary units).

4.2 Business Recommendations

1. Strengthen relationships with high-value customers

Northwind Traders should prioritise customer retention by maintaining strong relationships with its highest-spending customers. Personalised communication, customer satisfaction reviews, and targeted loyalty initiatives could help encourage repeat purchases.

2. Improve inventory planning

Products with consistently high sales volumes, such as Camembert Pierrot, should receive attention during inventory planning to reduce the risk of stockouts.

Lower-volume products should be evaluated using additional measures such as profitability, inventory turnover, and customer demand before making stocking decisions.

3. Investigate monthly sales fluctuations

Management should examine the factors contributing to revenue differences between months. Understanding whether changes are driven by seasonality, promotions, customer demand, or unusually large orders could improve forecasting and resource allocation.

4. Evaluate shipping costs and service performance

Although Speedy Express recorded the lowest average freight cost, management should compare shipping providers using delivery reliability, shipment characteristics, and customer satisfaction before making operational changes.

5. Monitor product profitability

Sales revenue and quantity alone do not determine profitability. Northwind Traders should incorporate product acquisition costs and other operating expenses into future analyses to identify products contributing the most profit.

4.3 Limitations of the Analysis

The analysis was based on the available historical Northwind Traders dataset.

The dataset did not provide all the information necessary to evaluate profitability, customer satisfaction, or delivery reliability comprehensively.

Consequently, recommendations involving pricing, inventory changes, and shipping providers should be validated using additional operational and financial data.

5. Project Conclusion

This project analysed Northwind Traders' historical sales data using SQL to identify patterns in revenue, customer purchasing behaviour, product performance, and shipping operations.

The analysis identified high-value customers, highlighted differences in product demand, examined monthly sales performance, and compared shipping costs across providers.

These findings provide a foundation for data-informed decisions relating to customer retention, inventory management, sales planning, and operational efficiency.

Technical Skills Demonstrated

Throughout the project, I applied:

- **SQL JOINs:** Combined related tables to analyse business transactions.

- **Aggregate Functions:** Used SUM(), AVG(), COUNT(), MAX(), and MIN() to calculate business metrics.
- **GROUP BY and HAVING:** Summarised transactional data and filtered grouped results.
- **Subqueries:** Compared individual records and aggregated results against calculated benchmarks.
- **Common Table Expressions (CTEs):** Structured multi-step queries for improved readability.
- **Date Functions:** Analysed monthly sales using SQLite's strftime() function.
- **Business Analysis:** Interpreted findings and developed recommendations based on the available evidence.

Future Improvements

Future work could extend this analysis by developing interactive dashboards in Power BI, examining customer purchasing frequency, investigating product profitability using additional cost data, and exploring sales forecasting.

This project demonstrates my ability to transform relational business data into meaningful insights using SQL.